

File Carving Analysis Report

Case: 14-822 — Host Based Forensics / File Carving Exercise

Field	Value
Client	Carnegie Mellon University (CMU)
Examiner	External IR Consultant
Report Date	2026-04-16 (UTC)
Evidence	~/cases/CORPUS/evidence/nps-2009-canon2-gen6.E01
Analysis Host	SANS SIFT Workstation

1. Evidence Integrity

Item	Value
Image format	EnCase 6 (EWF)
Acquisition date	2010-08-06 00:06:08 UTC
Acquiring OS	Darwin (macOS)
Media type	Removable disk (memory card)
Media size	31,129,600 bytes (29 MiB)
Sectors	60,800 @ 512 bytes/sector
MD5 (stored in E01)	750b509d8fbed37a5213480aaccfdc61
MD5 (verified at analysis)	750b509d8fbed37a5213480aaccfdc61
Integrity check	PASSED — image unmodified

2. Filesystem Overview

Item	Value
Partition table	MBR / DOS
Partition 1	Sectors 51–60799 (type 0x04 DOS FAT16, actual FAT12)
Filesystem	FAT12
Volume label	CANON_DC
OEM name	PwrShot
Cluster size	16,384 bytes (32 sectors)
Camera model	Canon PowerShot SD800 IS
Image resolution	640×480, 1600×1200, 2048×1536, 3072×2304
Focal length	4.6 mm (all frames, fixed zoom)

3. Filesystem Enumeration

fls -r -p -o 51 /mnt/ewf_cg6/ewf1

Category	Count
Allocated JPEG files (DCIM/100CANON/)	34
Deleted JPEG files (directory entry preserved)	3
Canon catalog file (M0100.CTG)	1
Total directory entries	38

3.1 Allocated Files

DCIM/100CANON/IMG_0003.JPG	(840,101 bytes)	inode 1031
DCIM/100CANON/IMG_0007.JPG	(865,313 bytes)	inode 1035
DCIM/100CANON/IMG_0009.JPG	(840,692 bytes)	inode 1037
DCIM/100CANON/IMG_0011.JPG	(771,052 bytes)	inode 1039
DCIM/100CANON/IMG_0013.JPG	(842,160 bytes)	inode 1041
DCIM/100CANON/IMG_0016.JPG	(853,839 bytes)	inode 1044
DCIM/100CANON/IMG_0017.JPG	(795,574 bytes)	inode 1045
DCIM/100CANON/IMG_0018.JPG	(784,455 bytes)	inode 1046
DCIM/100CANON/IMG_0019.JPG	(864,257 bytes)	inode 1047
DCIM/100CANON/IMG_0021.JPG	(819,599 bytes)	inode 1049

DCIM/100CANON/IMG_0022.JPG	(728,696 bytes)	inode 1050
DCIM/100CANON/IMG_0023.JPG	(858,798 bytes)	inode 1051
DCIM/100CANON/IMG_0024.JPG	(838,434 bytes)	inode 1052
DCIM/100CANON/IMG_0026.JPG	(768,385 bytes)	inode 1054
DCIM/100CANON/IMG_0027.JPG	(840,253 bytes)	inode 1055
DCIM/100CANON/IMG_0028.JPG	(815,636 bytes)	inode 1056
DCIM/100CANON/IMG_0029.JPG	(861,552 bytes)	inode 1057
DCIM/100CANON/IMG_0031.JPG	(749,202 bytes)	inode 1059
DCIM/100CANON/IMG_0032.JPG	(879,834 bytes)	inode 1060
DCIM/100CANON/IMG_0033.JPG	(845,375 bytes)	inode 1061
DCIM/100CANON/IMG_0034.JPG	(812,465 bytes)	inode 1062
DCIM/100CANON/IMG_0036.JPG	(882,337 bytes)	inode 1064
DCIM/100CANON/IMG_0038.JPG	(1,296,150 bytes)	inode 1038
DCIM/100CANON/IMG_0042.JPG	(1,361,807 bytes)	inode 1030
DCIM/100CANON/IMG_0043.JPG	(2,834,018 bytes)	inode 1032
DCIM/100CANON/IMG_0044.JPG	(105,195 bytes)	inode 1029
DCIM/100CANON/IMG_0045.JPG	(110,686 bytes)	inode 1033
DCIM/100CANON/IMG_0046.JPG	(116,135 bytes)	inode 1034
DCIM/100CANON/IMG_0047.JPG	(120,441 bytes)	inode 1036
DCIM/100CANON/IMG_0048.JPG	(124,906 bytes)	inode 1040
DCIM/100CANON/IMG_0049.JPG	(114,940 bytes)	inode 1042
DCIM/100CANON/IMG_0050.JPG	(127,191 bytes)	inode 1043
DCIM/100CANON/IMG_0051.JPG	(130,418 bytes)	inode 1048
DCIM/CANONMSC/M0100.CTG	(164 bytes)	inode 943621

3.2 Deleted Files (Directory Entry Present)

Filename	Inode	Size	Dir Entry	Timestamp	FAT Chain	Data Status
_MG_0025.JPG (IMG_0025)	1053	791,333 B	2008-12-23 14:14:40 UTC	Freed	Overwritten — unrecoverable	
_MG_0030.JPG (IMG_0030)	1058	867,833 B	2008-12-23 14:15:02 UTC	Freed	Partially overwritten; embedded thumbnail recovered	
_MG_0035.JPG (IMG_0035)	1063	820,105 B	2008-12-23 14:15:28 UTC	Freed	Overwritten — unrecoverable	

Note on TSK naming: FAT deletion replaces the first character of the 8.3 filename with 0xE5; TSK renders this as `_`. The original filenames were `IMG_0025.JPG`, `IMG_0030.JPG`, `IMG_0035.JPG`.

4. Deleted File Recovery Analysis

4.1 inode 1053 — IMG_0025.JPG

- **Cluster range:** sectors 39,437–40,982 (partition-relative)
- **Raw sector content at start:** 7b f1 5c df 88 fc 43 a7 — no JPEG signature (FF D8 FF)
- **Assessment:** Entire data region was reallocated and overwritten by subsequent writes to the card.
- **Recovered:** NO (791,333 bytes extracted but not a valid JPEG; icat returns overwritten data)

4.2 inode 1058 — IMG_0030.JPG

- **Cluster range:** sectors 47,469–49,163 (partition-relative)
- **Raw sector content at start:** 49 e6 7c a3 ec ac 4e 5d — no JPEG signature
- **Assessment:** First ~622 KB overwritten. Embedded EXIF thumbnail (at byte offset 622,592 within the original file space) survives intact.
- **EXIF capture time:** 2008-12-23 14:26:13 UTC
- **Recovered:** PARTIAL — 8,808-byte embedded thumbnail extracted to `exports/files/IMG_0030_exif_thumb.JPG`

4.3 inode 1063 — IMG_0035.JPG

- **Cluster range:** sectors 55,629–57,228 (partition-relative)
- **Raw sector content at start:** 9f 08 78 cf c0 5f 00 7e — no JPEG signature
- **Assessment:** Entire data region overwritten.
- **Recovered:** NO

5. Unallocated Space Carving

Three JPEG files were found in unallocated space with **no corresponding directory entries** — their directory entries were overwritten (deeper deletion than the three above). Their data blocks partially survived.

Carved File	Start Sector (fs)	Carved Size	EXIF Capture Time	Resolution	Status
CARVED_unalloc_sec21421.JPG	21,421	873,437 bytes	2008-12-23 14:13:45 UTC	1600×1200	FULLY RECOVERED
CARVED_unalloc_sec23533.JPG	23,533	9,164 bytes	2008-12-23 14:22:54 UTC	2048×1536	Thumbnail only (truncated)
CARVED_unalloc_sec32845.JPG	32,845	9,082 bytes	2008-12-23 14:23:01 UTC	2048×1536	Thumbnail only (truncated)

- **sec21421:** Full JPEG from 1728-sector unallocated gap; all data intact, EOI found at byte 873,435. This is a completely recoverable original photograph.
- **sec23533 & sec32845:** Only the leading EXIF APP1 block (including thumbnail) survived. The main image data was overwritten.

6. File Manifest (Exported Evidence)

File	Size (bytes)	MD5	Notes
CARVED_unalloc_sec21421.JPG	873,437	2c11f3b009dacc24896c0c0cb0df924c	Full JPEG, captured 2008-12-23 14:13:45
CARVED_unalloc_sec23533.JPG	9,164	aee494b6fe449c22c0d20fb2fe51e5b0	EXIF thumbnail only, 2008-12-23 14:22:54
CARVED_unalloc_sec32845.JPG	9,082	7de378d819b17222d0f40b69486d5207	EXIF thumbnail only, 2008-12-23 14:23:01
IMG_0030_exif_thumb.JPG	8,808	e02e1d4ce1baae6af98efb92850d0d87	Embedded thumbnail from deleted _MG_0030
M0100.CTG	164	b8ae356e040f8242122537c66e4dc961	Canon image catalog file
RECOVERED_IMG_0025.JPG	791,333	9ab4246657429642b4ee4a3a132036bc	icat output — NOT a valid JPEG (overwritten)
RECOVERED_IMG_0030.JPG	867,833	7b5bbe656fea8bfde21f162b2f150ed0	icat output — NOT a valid JPEG (first 622KB overwritten)
RECOVERED_IMG_0035.JPG	820,105	648a37e0f4c8a7a4eaaa7779186e776b	icat output — NOT a valid JPEG (overwritten)

7. EXIF Timeline — All Detected Images (Chronological)

All images taken with **Canon PowerShot SD800 IS**, focal length 4.6 mm. No GPS data in any frame.

Session A — 2008-12-23, 14:12–14:15 UTC (1600×1200, outdoor/daylight)

Time (UTC)	FS Sector	Status	Exposure
14:12:52	3,533	Allocated (IMG_0003 range)	1/80 s
14:13:12	10,093	Allocated	1/100 s
14:13:22	13,261	Allocated	1/100 s
14:13:31	16,621	Allocated	1/100 s
14:13:41	19,757	Allocated	1/100 s
14:13:45	21,421	CARVED (ghost file — no dir entry)	1/100 s
14:13:56	24,781	Allocated	1/100 s
14:14:01	26,477	Allocated	1/100 s
14:14:06	28,045	Allocated	1/100 s
14:14:10	29,581	Allocated	1/100 s
14:14:21	33,005	Allocated	1/100 s
14:14:26	34,637	Allocated	1/100 s
14:14:31	36,077	Allocated	1/100 s
14:14:36	37,773	Allocated	1/100 s
(14:14:40)	39,437	Deleted — IMG_0025 dir entry only	—
14:14:46	41,005	Allocated (IMG_0026)	1/100 s
14:14:50	42,509	Allocated	1/100 s
14:14:54	44,173	Allocated	1/100 s

Time (UTC)	FS Sector	Status	Exposure
14:14:58	45,773	Allocated	1/100 s
(14:15:02)	47,469	Deleted — IMG_0030 dir entry only	—
14:15:08	49,165	Allocated	1/100 s
14:15:14	50,637	Allocated	1/100 s
14:15:18	52,365	Allocated	1/100 s
14:15:23	54,029	Allocated	1/100 s
(14:15:28)	55,629	Deleted — IMG_0035 dir entry only	—
14:15:36	57,261	Allocated	1/100 s

Session B — 2008-12-23, 14:22–14:30 UTC (higher-resolution shots)

Time (UTC)	FS Sector	Status	Resolution	Exposure
14:22:46	8,077	Allocated	2048×1536	1/80 s
14:22:54	23,533	CARVED (ghost file — thumbnail only)	2048×1536	1/100 s
14:23:01	32,845	CARVED (ghost file — thumbnail only)	2048×1536	1/80 s
(14:26:13)	(48,685)	Embedded thumb from deleted IMG_0030	2048×1536	1/100 s
14:28:08	237	Allocated	2048×1536	1/100 s
14:30:39	2,989	Allocated	3072×2304	1/100 s

Session C — 2008-12-24, 20:21–20:22 UTC (640×480, low-light, evening)

Time (UTC)	FS Sector	Status	Exposure
20:21:45	173	Allocated	1/13 s
20:21:57	12,461	Allocated	1/20 s
20:22:10	12,685	Allocated	1/25 s
20:22:16	12,941	Allocated	1/25 s
20:22:22	13,197	Allocated	1/25 s
20:22:28	18,349	Allocated	1/25 s
20:22:36	18,605	Allocated	1/30 s
20:22:42	18,861	Allocated	1/30 s

Session C (640×480, slow shutter, evening of 2008-12-24) is a distinct shooting session. The low resolution and long exposure times are consistent with indoor/low-light conditions, approximately 30 hours after Sessions A/B.

8. Missing Image Numbering Analysis

The card contained images numbered up to IMG_0051. Significant gaps exist where files were fully deleted (no directory entry, no recoverable data):

Missing Images	Count
IMG_0001, 0002, 0004, 0005, 0006, 0008, 0010, 0012, 0014, 0015	10
IMG_0020	1 (ghost file at sector 21421 — RECOVERED)
IMG_0039, 0040, 0041	3
Total missing from directory	14

Of the 14 missing images: - **1 fully recovered** from unallocated space (sector 21421 → likely IMG_0020, 14:13:45) - **2 thumbnail-only** from unallocated space (sectors 23533, 32845) - **11 completely unrecoverable** — no data or directory evidence remains

9. Bulk Extractor Findings

bulk_extractor scanned the full 29 MiB image. Results: - **Email addresses:** 0 - **URLs / domains:** Minimal (no meaningful hits; no browsing history) - **GPS coordinates:** None in any EXIF record - **Credit card numbers:** None - **JPEG carved files:** 113 JPEG segment markers identified (aligned with files above) - **KML file:** Generated (exports/carved/kml.txt) — empty (no GPS data) - **EXIF records:** 37 (documented above)

10. Key Findings Summary

1. **Device:** Canon PowerShot SD800 IS compact camera, FAT12 memory card (29 MiB, volume CANON_DC).

2. **Image integrity verified.** MD5 hash matches acquisition record.
 3. **34 allocated JPEG files** remain on the card, captured across two days (2008-12-23 and 2008-12-24).
 4. **3 deleted files have surviving directory entries** (IMG_0025, IMG_0030, IMG_0035) but all data blocks were overwritten by subsequent card usage. Only IMG_0030's embedded EXIF thumbnail (8,808 bytes) was recovered.
 5. **1 additional deleted file was fully recovered** from unallocated space at sector 21421 (873,437 bytes, valid JPEG, captured 2008-12-23 14:13:45 UTC). This image has no directory entry — the directory entry was also deleted.
 6. **2 more ghost-file thumbnails** recovered from unallocated sectors 23533 and 32845 (EXIF portions only, ~9 KB each; captured 14:22:54 and 14:23:01 UTC on 2008-12-23).
 7. **No GPS metadata** in any image. No email, URL, or network artifacts.
 8. **Deletion pattern:** Consecutive images in the shooting sequence were selectively deleted (IMG_0025, IMG_0030, IMG_0035 deleted from an otherwise intact run; earlier images 0001–0015 and 0039–0041 were deleted at some earlier point with directory entries also removed). The selective deletion of every 5th image in a sequence (0025, 0030, 0035) may be significant.
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11. Output Artifacts

Path	Description
analysis/fls_output.txt	Full recursive file listing (allocated + deleted)
analysis/bodyfile.txt	MAC time bodyfile for timeline analysis
exports/fs_timeline.csv	Filesystem MAC timeline (UTC)
exports/files/	Extracted and carved image files (see Section 6)
exports/carved/	bulk_extractor feature files

Report prepared in accordance with DFIR chain-of-custody requirements. Evidence files were not modified. All analysis performed on read-only mounted image.